

"We Are Just One Step Away"; Embracing '4IR' With Realistic Transitional Framework Rooted With the Manufacturing Practices of RMG Industries in Bangladesh

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Abstract

Ready-made garment (RMG) industries of Bangladesh are dallying to accept the industrial culture of using '4IR' technologies and losing its competitive ground in the market. Consequently, it's enduring the economic stress of managing RMG-driven financial infrastructure. We conducted a mixed-method study involving 75 participants from 4 different RMG factories. Our findings unveil the ineptitude of the employers in evaluating system's capacity through operating manners, machines and automations in manufacturing to evolve with '4IR'. Whereas, we found equivocal appearance regarding technological exposures covered by numerous insecurities related to transitional decisions and job certainty to embrace '4IR'. We highlighted the stimulating factors for commencing '4IR' technology adoption through a flexible transitional framework of digitization. It is to emerge smarter industrial practices ensuring maximum utilisation of their resources and human engagement to enhance the country's economic landscape that's also essential for the growth of other RMG-based resource-constrained countries.

CCS Concepts

• **Human-centered computing, Human computer interaction (HCI), Empirical studies in HCI;**

Keywords

4IR, Ready-made Garment, RMG, Transitional Framework, Manufacturing, Technology Adoption.

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1 Introduction

The concept of '4IR' has been termed in many ways in different industries based on the fundamental theory of advancing technologies. The German federal government has named it 'High-Tech Strategy' [28]. Accomplishing the previous phases of the revolutions (mechanization, mass production & automation), industry 4.0 appeared with the phrase 'Internet of things and services' (IOT) [1]. In the developed countries, studies are carrying on with the manufacturing industries that have emerged their systems with IOT [47]. '4IR' is built on the following technology pillars- Big Data and AI analytics, Horizontal and vertical integration, Cloud computing, Augmented reality (AR), Industrial Internet of Things (IIOT), Additive manufacturing/3D printing, Autonomous robots, Simulation or digital twins and Cybersecurity [20]. However, in this research, we are exploring the penetration of relevant 4IR technologies into the production phases of RMG industries in Bangladesh. The implication of Big Data and Cloud Computing in the RMG industries lies in real-life data processing, spotting inefficiencies in different operations, production speed in different phases, scheduling, resource management, minimizing delays etc. Moreover, 3D printing technologies enable intricate design processing, 3D shape incorporation into fabrics etc. Besides, IOT tools including RFID can ensure sensor based tracking, monitoring, and detecting defects. Furthermore, AI and autonomous robots can certainly contribute to making customised patterns, cutting different types of fabrics precisely. In advanced stage, AI and robotic arms can handle complex stitching functions, transporting materials from one section to another, collaborating with vision to detect defects in involved phases of production. These technologies can eventually reduce the downtime and can ensure uninterrupted production phases, multiplying the production capacity and efficiency ensuring flexibility. Although the RMG industry owners are concerned and getting the urge to incorporate the technological applications of 4IR, there are diverse challenges around the industries. The knowledge of dealing with artificial intelligence (AI), robotics is atypical; skilling people in transitioning towards the smart factory is crucial for adaptability [13]. Besides technological innovation, organizational preparedness in managing the transformation [3] has been emphasized for synchronizing different machines, devices and other components to communicate thoroughly [27]. The shift is to introduce new operating systems, automation in production, horizontal and vertical system integration [38], combined actions of all [20], dynamic

balance of both technical skills and adaptability [41]. However, standardization of the system is also important for smooth data flow, secured connectivity [70] to ensure competitiveness, flexibility of adopting technology and continuous learning.

Relevant studies have already shown that despite having the potential of digitizing manufacturing systems, the scarcity of understanding the digital tools, trends, mindsets, agility [14, 48] is restricting this transition. Also, it has been marked as the operational necessity [62] that can collaboratively resolve social and environmental challenges [11] including establishing the sustainable development goals [53]. Different transitional frameworks [36], converting the system from lean to 4IR, cloud computing, Artificial Intelligence (AI) [24, 65, 68]. Unfortunately, RMG industry owners become directionless accumulating the production process with current policies [44]; the intervention of the technologies goes wrong [32] and interruptions occur (manual, semi-automated and automated). Since the age of revolution demands global competitiveness, customization, smart production system [34, 52], formation of a flexible schematic framework triggered this study exploring the following research questions-

RQ1: What is the scenario of encompassing automation into the manufacturing phases of RMG industries?

RQ2: What about the 4IR technology familiarity along with the workforce engagement throughout the manufacturing system?

To explore these RQs, we conducted Focus Group Discussions (FGD), in-person interviews of 75 participants from four RMG factories. Thematic analysis of qualitative data revealed interruptions in manufacturing, the tainted views of people etc. Furthermore, the overall familiarity of 4IR tools, upgraded machines and the contamination level have been highlighted through data visualisation from these four factories. Here, we contribute to the hitherto research in 4IR drawing an amenable skeleton to address the digital transition for RMG. It further contributes to the HCI and CSCW community through a collaborative in-house top-down model of industrial digitization to incorporate '4IR' technology which can be utilized in different contexts with small modifications.

2 Related Work

The researchers have focused on the '4IR' revolution, its appearance into the RMG sector, incorporating the individual components and technology interaction in the industries. This section presents the literature around the implementation of '4IR', readiness levels, impediments and transitional shifts throughout the RMG industries.

The RMG industry's technology adoption and integration characteristics have been explored by the researchers exploring readiness factors of the RMGs to adopt technology [4, 39], maturity levels of industry [21, 63], appropriate patterns of implementation and impediments [18]. The dimensions to ease the technology implementation process have been sorted out based on criteria to assess and improve maturity level of the organization where multiple methods have been applied [37, 49]. Categorization of the technologies and methods to adopt changes [54], ranking the elements based on the weights to ease the decision-making process [5] and enhancement of the sustainable performance in the RMG have been the focal points for the researchers to facilitate the adoption. It is primarily affected by industry awareness, government support, and

stakeholder exposure [6]. Also, the lack of technical expertise of the employers and the lethargy to adopt the smarter version of leadership have hindered this collaboration [29, 55, 60, 66]. Competitive advantage of inexpensive labor, and financial limitations as significant reasons for lagging behind to embrace the era of 4IR with its advanced prospectus in Bangladesh [26, 31, 51]. Studies indicate that Bangladesh has not yet reached that state of incorporating the complex technology culture of '4IR' through the execution of the smart factory concept [40]. Bangladesh has a negative reputation for lower labor productivity and unskilled workers compared to other RMG producing countries [15, 33]. The financial insolvency and the massive competitive advantage of the cheap labor market of Bangladesh, the feasibility of complete automation in the RMG sector is challenged [42]. The use of AI has been noticed in different stages of production but posed job loss threats to the employees. Moreover, the whole process of manufacturing has not been covered with a sequential automation process causing interruption [66]. The smart business or managerial concepts should also be aligned together to step towards the revolution [42]. The extent of automation should also be estimated with the calculation of man-to-machine ratio (MMR), the higher the automation, the lesser the number of employees [66].

The researchers in this domain have repeatedly expressed their concerns regarding the impediments to the technology transition and the challenges of implementing advanced technology. Studies have discussed the impediments to incorporating advanced systems into the industries through various models [30, 35, 43, 45, 61, 71] like DEMETEL, TISM, AHP, GRA, BWM, ISM & MICMAC Analysis etc. Data accessibility and smart management of the RMGs remains a challenge according to the researchers where vast amounts of data should be disseminated [67] across the industries. Moreover, adopting innovative technologies is highly expensive which hinders RMG firms from investing in technologies [12]. Bangladesh is not ready in terms of having skilled human resources with technical and applied knowledge of AI and BDA and adoption will increase their per unit cost leading to unwillingness of the owners [17, 58].

Since the industry matured mainly based on continuous cheap labor availability with low technical skills, it did not attract technical graduates in sufficient numbers to serve the organization while adopting new technologies [10]. Lower educated employees unwillingness to receive technology training [10] can be one impediment. Due to the massive involvement of manufacturing technologies, the factory environment demands highly competitive employees, teamwork, and flexibility to technology shift [57]. But as job displacement or job loss is a hyped topic, in-house training facilities, mental ability enhancement, gap minimization between managers and employees are emphasized as factors for enhancing employees' capability of setting goals and for effective organizational planning [58]. It calls for customized training programs and continuous employee development for enhancing employees' technical skills. However, ensuring the effectiveness of the training or learning session is a challenging issue for the industry owners as well [2].

In this research, we aim to bring out the necessary steps that can address such contradictions that impede the technology transition in the Bangladeshi RMG industries.

3 Research Method

This research inquiry emphasized on decoding the manufacturing phases along with the mixed method approach that disclosed technological familiarity and the ultimate human engagement with machines portraying the preparedness of the entire system for a uniform industrial transformation. We conducted 75 interviews from four different RMG industries of Dhaka and it took place between November 2023 to January 2024.

Participant Recruitment Participants were recruited using the purposive sampling method [51]. The participants were selected according to their designations. We recruited n=75 participants in total from both the employer and employee levels illustrated in Figure 3 in Appendix C. This is to identify the diverse involvement and motivational patterns with positional differences that their 4IR technologies usage perception [7]. All the participant details have been shown in Appendix B.

Study Design and Moderation To understand current technology usage scenarios across the RMG factories, we designed a semi-structured questionnaire; Open-ended questions covered participants' backgrounds, experience, automation level, skills, and their troubles in operations; while Likert scale, and closed-ended questions unfolded the industrial support for the professionals. There were 54 in-person interviews and 2 focus group discussions (FGD) comprising 11 participants, and 10 participants respectively. FGD was preferred to portray the familiarity levels to technological exposures of the officials belonging to the same hierarchical positions; however, two FGDs were formed into 2 different factories, others could not be managed due to participants' time constraints and lack of human resource allocation with heavy workloads. The questionnaire and participant numbers from different levels have been summarised in Table 1. All the participant details have been shown in Appendix B.

Interview Data Analysis In these four factories, individual respondents and focus group members were native citizens, so the traditional Bengali language was used during the conversation to leverage common cultural ground [38]. Then interviews were translated and transcribed in English. Inductive coding was conducted on the transcriptions [67]. In the initial stage, there were 111 open codes generated from the interviews (General perception on current technologies in the manufacturing phase, Viability of '4IR' technologies in RMG of Bangladesh, Limitations to embrace '4IR' etc.). Finally, it ended with 3 major themes generated from multiple iterations and analysis of the data given by the participants: Core Acknowledgement on Industrial Automation, Compatibility of the People for 4IR Technology Adoption, Aspects of Human Engagement for Successful Implementation. The main phases and interphases of manufacturing have been demonstrated in Appendix E.

Research Ethics Prior to the sessions, recruitment contacts were verbally informed about the study's purpose and type of questions. Participants were made aware of their right to terminate the study at any point. Focus group participants were already familiar with each other as co-workers which helped with rapport and trust. Discussions were recorded upon participants' consent. All data were stored with access limited only to the researchers. Only participants' organizational designations are mentioned and others

are used as pseudonyms. All participants were orally thanked after the interviews.

Positionality The research was conducted by mixed-gender researchers, born and brought up in Bangladesh, belonging to different socio-economic backgrounds with several years of qualitative and quantitative research experiences in Bangladesh on technology adoption into different industries such as medical, banking industries, education, etc. The researchers' backgrounds, positions, and research exposures have impacted research lenses to conduct the research.

4 Findings

The comprehensive theme found in this study is, beyond all the typically appraised contexts, the asynchrony and interjection of multi-operational modes (manual, semi-manual, automated/machine-driven) along with modern technologies outlined the story with a number of insecurities about utilizing coinciding resources, reaping advantages of 4IR and job persistence.

4.1 Grounding with Smarter Drivers of Technology

The interactions with operating processes and machines have been discussed in this section. The gross findings segmented the conventional manufacturing processes into five main phases with different operating modes (manual/semi-manual/machine-driven) shown in Figure 1a. The overall discussion said, that although the advanced tools are taking place in the manufacturing process, the manual activities are more intense than the machine-driven tasks. Since maximum machine-driven operations are observed in the sewing section, these all are conventional ones; the processes have yet to become automated. Many new features of the machines have been introduced, however, human intervention is obvious to associate the processes. The common insights suspect the complete feasibility of automation here. The cyber-physical mechanisms are yet to be established.

"Automation is not that easy. Because fashion is one of the most unpredictable modes of culture. No fixed automation is feasible in the world of RMG; human intervention is ubiquitous." (P53, Managing Director, F4).

Although some names of the flexible machines used in the RMG industries of developed countries are known to upper-level officials, the efficiency of interacting with these machines is yet to be evaluated.

"Everyone is not aware of it. We are running with the traditional mechanisms, and conventional tools. Skilled professional hands are lacking to provide us pure insights here." -(P42, Assistant General Manager-Production, F3).

The recurring names of the 4IR technologies among the experimented four factories have been visualized through Figure 1b. Moreover, the operations that are considered 'semi-manual' are typically initiated by humans and processed by machines or vice versa. These require human hands in coordinating the procedures,

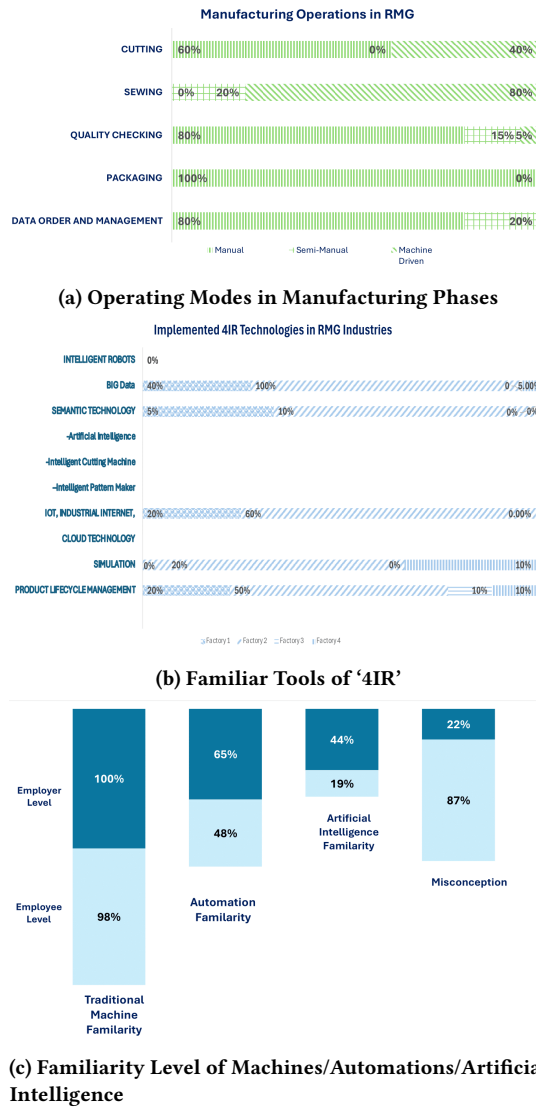


Figure 1: Technology scenario among our studied RMGs

causing inter-functional conflicts as well as uncertainty throughout the operating systems.

“After the advent of ERP software, many tasks are automated. Since we ourselves cannot design or functionally execute it, we buy specific packages where automation occurs intermittently. But the timing, layout, processing manner get interrupted.”- (P45, Assistant Manager- Production, F3)

The entire discussion factualized the familiarity of the 4IR tools and practical technological exposures into the systems. The stories shared by the participants suggested that their competence would increase with more exposure to the technological tools.

4.2 Human Interactions with New Technologies

In this segment of study, a distinct picture of applied engagement and familiarity to automation, AI have been captured in Figure 1c validating how real articulation is getting hampered due to the prevailing misconceptions. In case of automation and AI, the employer level was found mostly familiar due to their long term experiences and exposures of different industrial places and professionals. However, due to the lack of systematic disclosure and uncertainties about fabrics, fashions, it becomes difficult for them to fix their necessity, evaluating their capacity and compatibility of resources to incorporate 4IR tools.

“We purchased two automatic cutter machines. The layout of the floor has been altered for this. After installation, we noticed that a single machine could solely perform all our tasks and thus, another machine almost remained under utilization. There was a gap to evaluate our demand ” -(P42, Assistant General Manager- Production, F3)

In contrast, the acceptance of these tools are not very high in worker level which might be attributed to their unfamiliarity to advanced technologies and non-technical backgrounds of education. Many participants perceived the technologies like robots, AI and automation overwhelming for them, and considered it as a threat to their job.

“4IR can bring abundance to the RMG industries; however, people will lose their jobs. When robots replace human hands, everything will happen automatically. Factory will be empty, full of robots and machines.”- (P14, Floor in charge, F1)

The common view indicated that the digitization process is constrained due to the inconsistent and inaccurate perception presented above which lead to numerous assumptions that misguide both employers and employees to grasp the competitiveness of 4IR.

5 Discussion

This study explored the significance of enlightening the present insights regarding technological engagements into the manufacturing section of RMG plants so that an apprehensive framework can be suggested articulating the transitional activities of adopting ‘4IR’. Although the potential of 4IR [50, 64], its opportunities [25] and the impediments [40] have been explored in many concurrent studies, a basic transitional skeleton using existing technological and technical posture in the production process is yet to be disclosed for the RMG industries. This study puts into a distinct scenario regarding machines, and technologies; findings revealed cognizance level of basic 4IR technologies and extent of familiarity to the tradition of machines, automation, AI in manufacturing. Despite the limitations of establishing the 4IR concept [65], we highlighted the stimulating factors parameters for initiating the ‘4IR’ transitioning process evolving their present state of interactions with automation.

Since the demand of clothing has increasingly been triggered by the rapid growth of population, the present regime system of manufacturing in Bangladesh is incompetent with its traditional operations in production [23]. Researchers have found that incremental changes are more frequent than total reconfiguration [22].

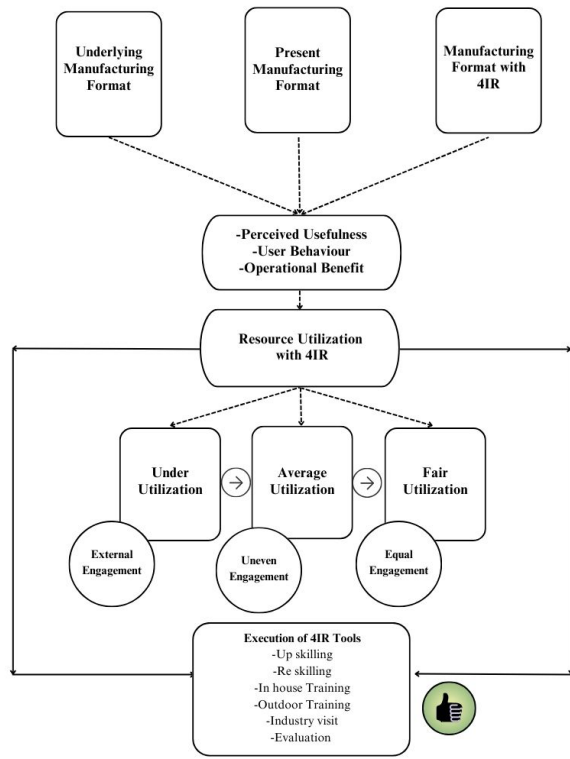


Figure 2: '4IR' Technology Transition Framework

In case of transition, socio-technical regime, niches, and socio-technical landscape [16] are required to be interacted properly. Different transitional schemes have been shared to be digitally affluent [19, 56, 59]. The inclusion of 4IR drivers have been regarded as core elements of industrial digitization [59]. However, the regime actors [8] along with prevailing practices can create deep structure [19, 69]. The dynamic and technical compatibility of the plants in dealing with the situations [9], operational capacity, and efficiencies [59] have been highlighted with perceived usefulness, user behavior, operational excellence, system quality, timeliness [46, 72] for digitization. According to our study findings and the concurrent literature, we, thereby, propose a '4IR' technology transitional framework (Figure 2) for the RMG industries of Bangladesh to start their conversion capturing their present strengths, opportunities, and shortfalls altogether.

Our suggested framework is focused on contemplating the major attributes related to the manufacturing practices, and associating technologies that are in use. The framework looks to identify the need of possible steps such as upskilling, reskilling, etc. based on the awareness and preparedness issues in the manufacturing stages of RMG industry. The execution of 4IR throughout the main manufacturing stages will be decided upon the capability of the RMG industries in identifying their resources for maximum utilization.

The framework will be segmented into three principal segments. Initial segment goes for the elementary manufacturing system analysis of the plant, the intermediate section stands for the evaluation

of cardinal manufacturing aspects and the final stage will solidify the execution of the latest tools of 4IR technologies complementing industries' present resources.

Segment 1: Elementary Manufacturing System Analysis

There are three clusters in this segment and every cluster will be analysed based on three pivotal criteria; perceived usefulness, user behavior and operational benefits.

Underlying Manufacturing Format: In this cluster, industry people will sort out the elements, practices and overall outcomes from the conventional format of operations to define their traditional prospectus of manufacturing that they have already experienced over the previous years. *Present Manufacturing Format:* In this part, current usage of upgraded manufacturing machines, modern tools and techniques, and behavioral feedback upon the unique exposure of new interventions will be analysed.

Manufacturing Format with 4IR: In this cluster, industrial professionals will be able to visualise the tools, mechanisms, machines and overall effects of cyber-physical systems into the manufacturing process that can be reaped dealing with the latest 4IR tools.

Since the mode of operations (manual, semi-manual, machine-driven) in each phase as we found in our study (Figure 1a), the number of machines, associating technologies, and the number of workers are fixed for a particular product manufacturing, the specification of some common attributes is required in this stage. These involve the number of operating modes implied in each phase, the number of machines running in the phase, the number of human hands engaged, time consumption in each phase, human-machine ratio, and efficiency across that ratio in each phase. It will eventually make them capable of comparing and justifying the perceived usefulness (user's belief on technologies that enhance their capability), user behaviors (actual overall usage of technologies that convenience them to upgrade), and operational benefits (accumulated results of beliefs and usage that make jobs efficient) among the old, present and upcoming '4IR' technologies in every immersed phase (Appendix D).

The compulsory evaluation of these clusters will lead RMG industries to segment 2 with the compilation of present status of resources based on employer-employee engagement with machines and relevant technological operations.

Segment 2: Utilization of Industrial Resources Depending on the employer-employee engagement ratio with the coinciding resources, this segment is identified as underutilized, average utilized, and fairly utilized status so that the industrial system can be leveled up from the deficient level to the superior one. During these inter-transitional expeditions, different modes of skill development, training facilities, consistent administration, process evaluation, and involved individuals will secure their strong stands gaining competitiveness in proper execution of 4IR technologies.

Under Utilisation: It will be termed while the machines, operations and technical interactions require external interventions beyond the expertise of employers and employees. They lack capability of maintaining and operating with the manufacturing systems tools. The requirements in this stage are basic skills development, smart interlinking approach development in employer-employee engagement, and opening the practical scope for exposures.

Since it involves external intervention of professionals, sequential exposures or scopes of the latest machines, automation, and AI

will be revealed, basic technological knowledge, planning, essential tools for setup, operating and managing skills will be spotted for the implied phases; thus, the uncertainty or suspicion of interacting with the system will be eliminated, advantageous aspects will be reaped, necessities will be met systematically and confidence level will be raised in both employer and employee levels to ease their tasks.

Average Utilization: It represents the uneven engagement of the employers and employees in dealing with the machines, operations and other technical manufacturing processes; either employers are proficient in interacting with the machines or technological tools or only the engineers/technicians are capable of performing with these. It demands smart collaboration with academia and leading industries, sharing knowledge and circulating ideas, and smart intervention of the human resources.

When technology management is partially undertaken by the employers and employees, knowledge circulates throughout the hierarchy levels; Academia-Industry collaboration distinguishes theoretical and real-life performances of 4IR technologies, automation, and AI. It eventually mitigates the level of misconceptions (Figure 1c), converting the myths to facts, and motivates them to prepare themselves with all their experiences, and practices over the years.

Fair Utilization: It is labeled when the industries can maintain the equal engagement of the employers and employees with the technological and technical tools throughout the manufacturing phases. This is regarded as the most superior context of utilizing their resources. The requirements in this stage are financial capacity analysis, smart leadership development and execution, and smart administration of the manufacturing processes.

In this scenario, the coordinated and integrated approaches of automation will be incorporated with the expertise of maintaining and operating these in both levels. The balanced shift in operating, managing, and monitoring the smart drivers of '4IR' will be established. The rationality will be developed throughout the employer and employee levels so that they can evaluate their competencies for surviving in the competitive world.

6 Conclusion

We presented this study focusing on the present 4IR tools engagement throughout the core manufacturing phases of RMG industries to develop a flexible and functional pathway for their digital conversion utilizing organizational resources so that the transitional initiatives can be taken promptly. We examined the functioning manufacturing modes, familiarity to the '4IR' technologies and their overall engagement with these. We described how the decision makers are muddled in evaluating their current resources, choosing the right direction to begin with their digital transformation. Also, the integrated approach of executing automation into the manufacturing sector is challenged due to the contamination about job loss entering into the age. The sequential approaches of evaluating their capacity in different levels and distinguishing their strengths, shortfalls and scopes according to their technological interactions have streamlined the framework to initiate the transitional journey. The combined engagement of the employer and employee levels can turn it into a natural substantial consequence

of execution. We also discussed the required linking attributes for the conditions and clusters so that the actions can be taken promptly evaluating requirements of 4IR technologies to digitise onwards. The industrial environment of any RMG plants can be transformed to the automated version for embracing '4IR' following the proposed framework. This study examined RMG manufacturing plants inside Dhaka only. Countrywide perspectives could make it more profound, accumulating broader participant groups. Our data visualization spotted their engagements with the technologies through different bar charts along with the qualitative analysis of their real story, hence inductive in nature including observer bias, self-censoring participants for sensitive topics, and drawbacks of the generalizability of results. Besides RMGs, studying other manufacturing industries could also involve multi-dimensional factors for digital transition in the developing countries that could verify the sensitivity of our proposed framework with both qualitative and quantitative analysis of their real scenario.

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A Appendix

The following Table 1 summarizes the questionnaire and participant numbers from different levels.

Table 1: Participant and Question Distribution

Total Number of Participants	75
Employer Level Participants	23
Employee Level Participants	52
Total Number of Questions	15
Open Ended Questions	10
Closed Ended Questions	3
Like-art Scale Questions	2

B Appendix

The following Tables 2 and 3 illustrate the participants' details e.g., factory, age, gender, Designation, experience, and educational level for the comprehensive understanding of the reader about the participants' context.

C Appendix

Figure 3 presents the designations of the recruited participants showing their representation as employer and employee segments used throughout the paper.

D Appendix

Table 4 lists the Industrial Terms Used in the Factory.

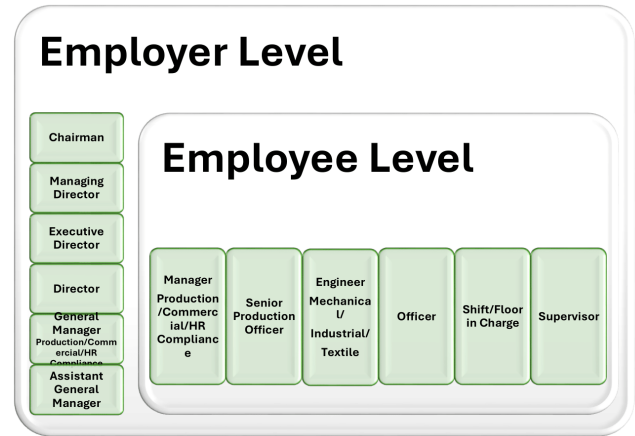


Figure 3: Designation of Participants

E Appendix

Table 5 presents the Operations Involved in Manufacturing.

Table 2: Participant Details Across the First 2 Factories (F1, F2)

Factory	ID	Age (Years)	Gender	Designation	Experience (Years)	Education Level
F1	P1	>50	Male	Managing Director	>30	Master's
F1	P2	>50	Male	Chairman	>30	Master's
F1	P3	>50	Male	Director	21-30	Master's
F1	P4	>50	Female	Executive Director	21-30	Master's
F1	P5	41-50	Male	General Manager-Production	21-30	Bachelor's
F1	P6	31-40	Male	Assistant General Manager-Commercial	21-30	Bachelor's
F1	P7	31-40	Male	Assistant General Manager-Human Resource	21-30	Bachelor's
F1	P8	31-40	Male	Assistant General Manager-Production	21-30	Bachelor's
F1	P9	18-30	Male	Textile Engineer	<10	Bachelor's
F1	P10	18-30	Female	Environment Engineer	<10	Bachelor's
F1	P11	31-40	Male	Manager-Production	10-20	Bachelor's
F1	P12	31-40	Male	Manager-Quality	10-20	Bachelor's
F1	P13	31-40	Male	Shift-In charge	10-20	Higher Secondary
F1	P14	18-30	Male	Floor-In charge	<10	Higher Secondary
F1	P15	18-30	Male	Floor Supervisor	<10	Higher Secondary
F2	P16	>50	Male	Executive Director	>30	Master's
F2	P17	>50	Male	Director-In charge	>30	Master's
F2	P18	>50	Male	Senior General Manager-Production	21-30	Master's
F2	P19	41-50	Female	General Manager-Production	21-30	Master's
F2	P20	18-30	Male	Senior Operations Instructor	21-30	Master's
F2	P21	31-40	Male	Assistant General Manager-Commercial	21-30	Master's
F2	P22	31-40	Male	Assistant General Manager-Compliance	21-30	Master's
F2	P23	31-40	Male	Assistant General Manager-Production	21-30	Bachelor's
F2	P24	31-40	Male	Manager-Production	21-30	Master's
F2	P25	31-40	Male	Manager-Quality	21-30	Bachelor's
F2	P26	31-40	Female	Manager-Human Resource	21-30	Bachelor's
F2	P27	18-30	Male	Assistant Manager-Quality	21-30	Bachelor's
F2	P28	31-40	Male	Assistant Manager-Production	21-30	Bachelor's
F2	P29	18-30	Male	Textile Engineer	<10	Bachelor's
F2	P30	31-40	Male	Mechanical Engineer	10-20	Bachelor's
F2	P31	18-30	Female	Environment Engineer	<10	Bachelor's
F2	P32	31-40	Male	Shift-In charge	10-20	Higher Secondary
F2	P33	18-30	Male	Floor-In charge	<10	Higher Secondary
F2	P34	18-30	Male	Floor Supervisor	<10	Higher Secondary

Table 3: Participant Details Across the Last 2 Factories (F3, F4)

Factory	ID	Age (Years)	Gender	Designation	Experience (Years)	Education Level
F3	P35	>50	Male	Managing Director	>30	Master's
F3	P36	>50	Male	Chairman	>30	Master's
F3	P37	>50	Female	Director-1	21-30	Master's
F3	P38	41-50	Male	Director-2	21-30	Bachelor's
F3	P39	41-50	Female	Executive Director	21-30	Bachelor's
F3	P40	31-40	Male	Assistant General Manager-Commercial	21-30	Bachelor's
F3	P41	31-40	Male	Assistant General Manager-Compliance	21-30	Bachelor's
F3	P42	31-40	Male	Assistant General Manager-Production	21-30	Bachelor's
F3	P43	41-50	Male	Manager-Human Resource	21-30	Bachelor's
F3	P44	31-40	Male	Assistant Manager-Quality	21-30	Bachelor's
F3	P45	31-40	Male	Assistant Manager-Production	21-30	Bachelor's
F3	P46	31-40	Male	Assistant Manager-Audit	21-30	Bachelor's
F3	P47	18-30	Male	Production Officer 1	10-20	Bachelor's
F3	P48	18-30	Male	Production Officer 2	10-20	Bachelor's
F3	P49	18-30	Male	Production Officer 3	10-20	Bachelor's
F3	P50	18-30	Male	Shift-In charge	<10	Bachelor's
F3	P51	18-30	Male	Floor-In charge	<10	Bachelor's
F3	P52	18-30	Male	Floor Supervisor	<10	Higher Secondary
F4	P53	>50	Male	Managing Director	>30	Master's
F4	P54	>50	Female	Chairman	>30	Master's
F4	P55	>50	Female	Director-1	21-30	Master's
F4	P56	>50	Male	Director-2	21-30	Bachelor's
F4	P57	41-50	Female	Executive Director	>30	Bachelor's
F4	P58	41-50	Male	General Manager-Commercial	10-20	Bachelor's
F4	P59	31-40	Male	Assistant General Manager-Commercial	10-20	Bachelor's
F4	P60	31-40	Male	Assistant General Manager-Compliance	21-30	Bachelor's
F4	P61	31-40	Male	Assistant General Manager-Production	21-30	Bachelor's
F4	P62	31-40	Male	Manager-Quality	21-30	Bachelor's
F4	P63	31-40	Male	Manager-Production	21-30	Bachelor's
F4	P64	31-40	Male	Manager-Human Resource	21-30	Bachelor's
F4	P65	41-50	Male	Assistant Manager-Quality	10-20	Bachelor's
F4	P66	31-40	Male	Assistant Manager-Production	10-20	Bachelor's
F4	P67	18-30	Male	Assistant Manager-Audit	10-20	Master's
F4	P68	18-30	Male	Production Officer 1	10-20	Bachelor's
F4	P69	18-30	Male	Production Officer 2	10-20	Bachelor's
F4	P70	18-30	Male	Shift-In charge 1	10-20	Bachelor's
F4	P71	18-30	Male	Shift-In charge 2	10-20	Bachelor's
F4	P72	18-30	Male	Floor-In charge 1	<10	Bachelor's
F4	P73	18-30	Male	Floor-In charge 2	<10	Higher Secondary
F4	P74	18-30	Male	Floor Supervisor 1	<10	Higher Secondary
F4	P75	18-30	Male	Floor Supervisor 2	<10	Higher Secondary

Table 4: Industrial Terms Used in the Factory

Factory Terms	Definitions
RMG Sector	Ready-made garments industry sector.
Standard Allowed Minute (SAM)	Standard allowed minute (SAM) or standard minute value (SMV) is used to measure the task or work content of a garment; basically, the time required for a qualified worker working at "Standard Performance" to perform a given task.
LEAN, 5S, 6 SIGMA	Lean - Lean manufacturing is a methodology that focuses on minimizing waste within manufacturing systems while simultaneously maximizing productivity. The strategy aims to increase efficiency by eliminating waste, optimizing processes, and cutting costs. 5S - The 5S methodology is summarized by the philosophy, "a place for everything and everything in its place." It originated from 5 Japanese words: • <i>Sort (Seiri)</i>: Eliminate that which is not needed. • <i>Straighten (Seiton)</i>: Organize what remains after sorting. • <i>Shine (Seisou)</i>: Clean and inspect the work area. • <i>Standardize (Seiketsu)</i>: Write standards for 5S. • <i>Sustain (Shitsuke)</i>: Consistently apply the 5S standards. 6 Sigma - Six Sigma is a quality control method aimed at significantly reducing defects and improving processes.
Semantic Technology	Semantic Technology defines and links data on the Web (or within an enterprise) by developing languages to express rich, self-describing interrelations of data in a form that machines can process.
IoT, Industrial Internet, Cloud Technologies, Artificial Intelligence (AI)	IoT - Refers to the network of connected devices that communicate with each other and the cloud. AI - A set of technologies that allow computers to perform tasks typically requiring human intelligence.
Product Lifecycle Management	Product lifecycle management (PLM) helps businesses plan and execute all stages of a product's life, from design and development to distribution, marketing, and sales.
Visual Computing	Visual computing covers computer science disciplines related to images and 3D models, including computer graphics, image processing, computer vision, and augmented reality.
Industrial Automation	Industrial automation uses control systems and robotics to automatically operate industrial processes and machinery, reducing the need for human intervention.
Intelligent Robots	Intelligent robots can learn and adapt to new situations using advanced AI systems, allowing autonomous decision-making based on environmental data.
Cybersecurity	Cybersecurity protects systems, networks, and programs from digital attacks aimed at accessing, altering, or destroying sensitive information, extorting money, or disrupting business operations.
Industrial Big Data	Industrial big data refers to massive data sets that are too large or complex for traditional processing tools, providing deeper insights but also posing risks of false discoveries.
Simulation	A simulation is an imitative representation of a real-world process or system, often used interchangeably with models for analysis and forecasting.
Perceived Usefulness	Reflects the user's belief that sums up whether the primitive format of technologies/machines/operating modes in each phase can enhance their capability of completing tasks.
User Behavior	Refers to actual and overall usage of previous/present machines, technologies, and involved modes that make them capable of assuming their situation with and without 4IR technologies and convenience them for upgradation.
Operational Benefit	Accumulated results of the perceived usefulness and behavior of the users that make their jobs efficient with willingness to adopt changes.

Table 5: Basic Operations and Their Sub-Sections

Sl. No.	Basic Operations	Sub-Sections
1	Cutting	1. Design Laying 2. Cutting 3. Numbering 4. Cut panel inspection 5. Recut 6. Fusing
2	Sewing (1)	Care Label Attach With Back Part Placket & Body Part Matching Placket Attach Positioning Marking Placket Rolling Placket Attach With Front Part Placket Edge Cut & Tack Half Moon Cut & Iron Main & Size Label Attach Half Moon Attach With Back Part Front And Back Part Matching Shoulder Joining Collar Servicing Collar Making Collar Attach With Tape Back Tape Top Stitch Lower Placket Close Upper Placket Top Stitch Placket Box Stitch Cuff Joining Cuff Top Stitch
3	Sewing (2)	Sleeve Pair Matching With Body Part Sleeve Joining Arm Hole Top Stitch Bottom Hem Stitch Side Split Tack Side Split Rolling & Top Stitch Side Seam Sleeve Close & Open Tack Button Hole Creating At Upper Placket Button Attach Positioning Mark At Lower Placket Button Attach With Lower Placket Button Push Final Thread Cut
4	Packaging	Get-up Check Measurement Check Re-iron Pack
5	Data and Order Management	Product Order Placement Product Tracking and Queue Maintenance Technology Monitoring and Management Inventory Checking and Management By-product Data Management